ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

Q1(b) [3 marks]

$$Sensitivity = 10mV/N$$

The ADC has 12 bits, so there are 4096 levels.

Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

Question 2 [10 marks]

Q2(a) [2 marks]

Choose $x_1 = \theta$ and $x_2 = \dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = [\tau - b\,x_2 - mg\ell \sin(x_1)]/J$. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since $|\sin \theta_e|$ less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to $mg\ell = 2.4525$ N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \leq \sin\theta_e \leq 1$, the required torque must lie between -2.4525 and $+2.4525$ N m.

Q2(b) [2 marks]

$$A = [[0, 1], [-mg\ell sin(60^\circ)/J, -b/J]] = [[0, 1], [-106.2, -2.5]],$$
$$B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is $K_p(\theta_d - \theta)$, which must balance $mg\ell \sin\theta$.

Q2(d) **[2 marks]**

Question 3 [10 marks]

Q3(a) **[2 marks]**

Q3(b) **[2 marks]**

Tustin uses s = (2/T)(z-1)/(z+1).

Q3(c) **[4 marks]**

The characteristic polynomial is $s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. I match it to $s^2+10s+100$ and get $\ell_1=8$, $\ell_2=84$.

Q3(d) **[2 marks]**

An observer uses the plant model and known control input to estimate unmeasured states such as velocity, and its poles can be designed. A simple low-pass filter cannot exploit the state-space model in this way. A filter may be safer when the model is poorly known or unreliable, because it relies less on model accuracy.

The observer uses the plant model and the measured output to reconstruct the unmeasured state. Its convergence depends on the model and the observer poles. A low-pass filter is simpler, but generally introduces phase lag and does not explicitly use the plant dynamics.

Question 4 [10 marks]

Q4(a) **[2 marks]**

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) **[2 marks]**

The nominal controller has unity DC gain, so $\theta ss=1$ and the tracking error is zero even after the actuator gain changes.

Q4(c) **[4 marks]**

I obtain $s^3+(a+bk_2)s^2+(c+bk_1)s-bK_I$. Desired polynomial is $s^3+8s^2+24s+32$, so $k_2=6$, $k_1=18$, K_I=-32.

Q4(d) **[2 marks]**